

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI, KARNATAKA- 590018**



A Project-Phase1 Report on

**“Multimodal IVF Embryo Implantation Prediction Using
Blastocyst Image Processing and Clinical Data Fusion”**

Submitted in partial fulfillment of the requirement for the award of the degree

**Bachelor of Engineering
in
ELECTRONICS AND COMMUNICATION ENGINEERING
Prescribed by
Visvesvaraya Technological University**

By

VIBHA I S	1BY23EC122
ASHITHA M	1BY23EC024
AKSHAYA RAMESH	1BY23EC009
SIREESHA T S	1BY23EC105

Under the guidance of

Prof.Chandrababha R
Assistant Professor, Dept. of ECE, BMSIT & M



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(An Autonomous Institute affiliated to Visvesvaraya Technological University, Belagavi)

Avalahalli, Yelahanka, Bengaluru-560119

2025-2026

BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT
(An Autonomous Institute affiliated to Visvesvaraya Technological University, Belagavi)
Yelahanka, Bengaluru-560119

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Institute Vision

To emerge as one of the finest technical institutions of higher learning, to develop engineering professionals who are technically competent, ethical and environment friendly for betterment of the society.

Institute Mission

Accomplish stimulating learning environment through high quality academic instruction, innovation and industry-institute interface.

Department Vision

Be a pioneer in providing quality education in electronics, communication, and allied engineering fields to serve as a valuable resource for industry and society

Department Mission

To nurture students with a strong foundation in theoretical and practical concepts through innovative pedagogy and industry interface. To foster interdisciplinary research, entrepreneurship skills and instill professional ethics by providing experiential learning opportunities.

Program Educational Objectives (PEOs):

After three to four years of graduation our graduates will:

PEO 1: Establish successful careers in electronics, communication, and allied engineering domains.

PEO 2: Pursue advanced studies and conduct research to devise innovative solutions, propelling technological advancements.

PEO 3: Demonstrate leadership, effective communication, and teamwork while upholding ethical practices through sustainable engineering solutions.

Program Specific Outcomes (PSOs)

PSO1: Design and analyze digital and analog circuits using industry-standard Electronic Design Automation (EDA) tools, applying electronic fundamental principles for VLSI system implementation.

PSO2: Design and develop communication and signal processing systems using fundamental principles of analog and digital communication, effectively utilizing analytical tools and techniques.

PSO3: Integrate, debug, and deploy embedded hardware and software solutions for various real-time applications, demonstrating proficiency in programming, interfacing, and system-level design.

Program Outcomes (POs)

- PO1** Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- PO2** Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- PO3** Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- PO4** Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5** Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- PO6** The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)
- PO7** Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- PO8** Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- PO9** Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10** Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- PO11** Life-Long Learning: Recognize the need for, and have the preparation and ability for
 - i) independent and life-long learning
 - ii) adaptability to new and emerging technologies
 - and iii) critical thinking in the broadest context of technological change. (WK8)

Knowledge and Attitude Profile (WK)

- WK1** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- WK2** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- WK3** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- WK4** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- WK5** Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- WK6** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- WK7** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
- WK8** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
- WK9** Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(An Autonomous Institute affiliated to Visvesvaraya Technological University, Belagavi)

Avalahalli, Yelahanka, Bengaluru-560119



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

Certified that the project work entitled “**Multimodal IVF Embryo Implantation Prediction Using Blastocyst Image Processing and Clinical Data Fusion**” carried out by **Ms. VIBHA I S (1BY23EC122)**, **Ms. ASHITHA M (1BY23EC024)**, **Ms. AKSHAYA RAMESH (1BY23EC009)**, and **Ms. SIREESHA T S (1BY23EC105)** bonafide students of **BMS Institute of Technology and Management** in partial fulfilment for the award of **Bachelor of Engineering in Electronics and Communication Engineering** prescribed by **Visvesvaraya Technological University, Belagavi** during the academic year **2025-26**. It is suggested that all corrections/suggestions indicated for internal assessment have been shown in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said degree.

Signature of Guide/s
Prof. Chandrabha R

Signature of HOD
Dr. A SHOBHA RANI

Signature of Principal
Dr. Sanjay H. A

ABSTRACT

Accurate prediction of embryo implantation success in In Vitro Fertilization (IVF) remains a significant challenge in assisted reproductive technology. This study presents an optimized machine learning framework for binary classification of IVF implantation outcomes using fused clinical and embryo-derived features. Following preprocessing and feature selection, class imbalance was addressed using Synthetic Minority Oversampling Technique (SMOTE) with an 0.85 sampling ratio. Feature normalization was performed using StandardScaler to improve model stability and convergence.

An optimized Extreme Gradient Boosting (XGBoost) classifier was developed with tuned hyperparameters including controlled tree depth, subsampling, regularization, and adaptive learning rate to minimize overfitting while improving generalization. Model performance was further enhanced through threshold optimization based on a weighted combination of accuracy, F1-score, and recall. Evaluation was conducted on separate validation and test datasets using metrics including accuracy, precision, recall, specificity, F1-score, and ROC-AUC.

The proposed model achieved strong predictive performance, demonstrating the effectiveness of ensemble boosting techniques for IVF outcome prediction. Results indicate that balanced learning, feature scaling, and threshold optimization significantly improve implantation prediction reliability. The developed pipeline provides a scalable and reproducible framework suitable for future deployment in clinical decision-support systems and FPGA-based healthcare acceleration platforms.

Keywords: assisted reproductive technology clinical data analysis,, embryo implantation prediction, embryo classification, feature selection, IVF, machine learning, SMOTE, XGBoost,

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible and whose constant guidance and acknowledgement crowned our effort with success.

We express our profound gratitude to our Principal, **Dr. Sanjay H. A**, BMS Institute of Technology and Management for providing all the facilities and encouragement.

We thank our HOD, **Dr. A. Shobha Rani**, for the inspiration, guidance and valuable suggestions.

We sincerely thank our Project coordinators, **Dr. Thejaswini S(Associate Professor)** and **Prof. Vivekanand Upadhye(Assistant Professor)** for their valuable time, suggestions, and technical support in conducting the project presentation and writing the report.

Our sincere gratitude to our guide, **Prof. ChandraPrabha R(Assistant Professor)** for her valuable time, patience, suggestions and periodic evaluation that was conducive for the project.

We would also like to thank all the teaching and non-teaching staff of the Department of Electronics and Communication Engineering for their cooperation and motivation.

Finally, we express our cordial thanks to our parents and friends for their support and guidance throughout the project preparation.

VIBHA I S (1BY23EC122)

ASHITHA M(1BY23EC024)

AKSHAYA RAMESH (1BY23EC009)

SIREESHA T S (1BY23EC105)

CONTENTS

<i>Abstract</i>	<i>i</i>
<i>Acknowledgement</i>	<i>ii</i>
<i>Contents</i>	<i>iii</i>
<i>List of Figures</i>	<i>iv</i>
<i>List of Tables</i>	<i>v</i>

Chapter	Page No.
Chapter 1: INTRODUCTION	
1.1 Overview	1
1.2 Summary	2
Chapter 2: LITERATURE SURVEY	
2.1 Review	3-6
2.2 Motivation	6
2.3 Research Gaps and Challenges	6-7
2.4 Statement of the Project	7-8
2.5 Objectives	8
2.6 Methodology	9-10
2.7 Summary	10
Chapter 3: DESIGN AND IMPLEMENTATION	
3.1 Introduction	11
3.2 Implementation	12-17
3.3 Results and discussion	17-18
3.4 Summary	18
Chapter 4: CONCLUSION	
4.1 Conclusion	19
4.2 Advantages	19-20
4.3 Disadvantages	20

4.4 Future Scope

20-21

REFERENCES

22-23

STUDENTS PROFILE

APPENDIX:

Code used in the project

LIST OF FIGURES

Fig. 3.1: Block diagram of proposed system	11
Fig. 3.2: Sample embryo image before and after clahe enhancement	12

LIST OF TABLES

Table 3.1: GLCM texture features extracted per image	13
Table 3.2: XGBoost hyper parameter configuration	16-17
Table 3.3: Model performance on validation and test sets	17-18

CHAPTER 1

INTRODUCTION

1.1 Overview

In Vitro Fertilization (IVF) is one of the most widely used assisted reproductive technologies (ART) globally, offering hope to millions of couples facing infertility challenges. The IVF process involves stimulating the ovaries to produce multiple eggs, retrieving those eggs, fertilizing them in a laboratory setting, and transferring the resulting embryos into the uterus with the goal of achieving a successful pregnancy.

Despite significant advances in reproductive medicine, the success rate of IVF cycles remains modest, typically ranging from 30% to 50% depending on patient age and clinical factors. A critical determinant of IVF success is embryo implantation — the process by which a fertilized embryo attaches to the uterine lining and begins to develop. Predicting which embryos have the highest implantation potential is a central challenge for embryologists and clinicians.

Traditionally, embryo selection has relied on morphological grading systems, where trained embryologists visually assess embryo quality based on characteristics such as cell number, symmetry, and fragmentation. While useful, these assessments are inherently subjective and suffer from inter-observer variability. The emergence of time-lapse imaging, blastocyst culture systems, and molecular techniques such as preimplantation genetic testing (PGT) have improved selection accuracy but remain expensive, time-consuming, and not universally accessible.

Machine learning (ML) and deep learning approaches have emerged as powerful tools for objective embryo assessment. These methods can extract and analyze complex patterns from embryo images and clinical data that may be imperceptible to the human eye, thereby providing more consistent and potentially more accurate predictions of implantation outcomes.

This project presents a multimodal machine learning framework that integrates image-derived texture features extracted from blastocyst images with structured clinical IVF parameters. The pipeline incorporates CLAHE-enhanced image preprocessing, GLCM-based texture feature extraction, clinical feature fusion, SMOTE-based class balancing, and an optimized XGBoost

classifier with threshold tuning to deliver a robust binary classification of IVF implantation outcomes.

1.2 Summary

This chapter introduced the clinical context of IVF and embryo implantation prediction. The limitations of traditional morphological assessment were highlighted, motivating the need for an objective, data-driven approach. The proposed framework combines blastocyst image texture analysis with clinical parameter integration through a structured machine learning pipeline. The following chapters provide a comprehensive review of related literature, describe the design and implementation of the system, and present the experimental results along with conclusions and future directions.

CHAPTER 2

LITERATURE SURVEY

2.1 Review

The field of IVF outcome prediction has seen significant advances in recent years, with researchers applying increasingly sophisticated machine learning and deep learning techniques to improve prediction accuracy and clinical utility. The following works represent key contributions from 2020 to 2026.

1. Multimodal Intelligent Prediction Model for In Vitro Fertilization (2026)

- a) **Method:** VaTEP multimodal deep learning model combining 3D CNN, Temporal Transformer, and cross-attention fusion.
- b) **Data/Hardware:** 9,786 IVF patient records with embryo time-lapse videos and 39 clinical variables.
- c) **Result:** Improved prediction of fetal heartbeat, pregnancy type, and miscarriage versus live birth through multimodal integration.
- d) **Limitation:** Dataset access is restricted due to privacy concerns; retrospective hospital data may limit generalization.

2. Evaluating the Concordance Between AI-Based and Conventional Embryo Selection (2026)

- a) **Method:** Multicenter retrospective analysis comparing the ESA AI algorithm with 20 expert embryologists.
- b) **Data/Hardware:** 1,681 embryo pairs collected from clinics across 5 countries.
- c) **Result:** AI achieved 70.1% accuracy and outperformed 14 out of 20 embryologists.
- d) **Limitation:** Requires larger prospective multicenter trials before clinical deployment.

3. A Comprehensive Survey on Deep Learning Techniques for Embryo Quality Assessment (2025)

- a) **Method:** Review of CNNs, GANs, and Transformer-based embryo assessment models.
- b) **Data/Hardware:** Public and private embryo imaging datasets.
- c) **Result:** Transformers and GANs show strong potential for reducing subjectivity in embryo grading.

- d) **Limitation:** Highlights black-box nature of deep learning, data imbalance, and poor cross-clinic generalization.

4. Machine Learning-Based Prediction of IVF Outcomes: The Central Role of Female Preprocedural Factors (2025)

- a) **Method:** Random Forest, Logistic Regression, and Gradient Boosting models trained on clinical IVF data.
- b) **Data/Hardware:** Dataset containing age, AMH level, BMI, infertility duration, and embryo quality.
- c) **Result:** Female pre-procedural factors identified as major predictors of IVF success.
- d) **Limitation:** Limited clinical centers and absence of genetic/lifestyle data.

5. Machine Learning Based Assessment and Predictive Analysis of In-Vitro Fertilization Success Rate (2024)

- a) **Method:** Machine learning classifiers using clinical and demographic IVF datasets.
- b) **Data/Hardware:** Hormone levels, infertility causes, and embryo parameters.
- c) **Result:** ML models effectively estimate IVF success probability.
- d) **Limitation:** Small dataset and lack of real-world clinical validation.

6. Catalyzing IVF Outcome Prediction: Exploring Advanced Machine Learning Paradigms (2024)

- a) **Method:** Ensemble learning models such as LogitBoost applied to IVF clinical data.
- b) **Data/Hardware:** Patient demographic and treatment datasets.
- c) **Result:** Achieved prediction accuracy of nearly 96% for IVF success prediction.
- d) **Limitation:** Requires external validation and larger datasets.

7. Comparative Study of Machine Learning Approaches Integrated with Genetic Algorithm for IVF Success Prediction (2024)

- a) **Method:** RF, ANN, SVM, RPART, and AdaBoost combined with Genetic Algorithm feature selection.
- b) **Data/Hardware:** Clinical IVF datasets.
- c) **Result:** AdaBoost + GA achieved highest prediction accuracy of approximately 89.8%.
- d) **Limitation:** Dataset source unclear and embryo imaging information excluded.

8. Predictors of Success Following IVF (2024)

- a) **Method:** Review of 43 studies using statistical and ML approaches.
- b) **Data/Hardware:** Multiple IVF clinical datasets from previous studies.
- c) **Result:** Identified age, BMI, embryo quality, AFC, and oocyte count as major predictors.
- d) **Limitation:** Review-based work without development of a new predictive model.

9. Embryo Selection Through Artificial Intelligence Versus Embryologists: A Systematic Review (2023)

- a) **Method:** Systematic review of AI-based embryo image analysis and clinical prediction systems.
- b) **Data/Hardware:** Embryo imaging and IVF clinical datasets from multiple studies.
- c) **Result:** AI systems often outperform embryologists in predicting embryo viability.
- d) **Limitation:** Lack of standardized datasets and prospective validation.

10. Comparison of Machine Learning Classification Techniques to Predict Implantation Success in an IVF Treatment Cycle (2022)

- a) **Method:** Random Forest and Super Learner classification models.
- b) **Data/Hardware:** Dataset of 939 embryo transfer cycles.
- c) **Result:** Random Forest and Super Learner produced the best implantation prediction performance.
- d) **Limitation:** Retrospective dataset from a limited cohort reduces generalizability.

11. Individualized Embryo Selection Strategy Developed by Stacking Machine Learning Model for Better IVF Outcomes (2021)

- a) **Method:** Stacking ML model using XGBoost for personalized embryo selection.
- b) **Data/Hardware:** 9,211 IVF patients and 10,076 embryos.
- c) **Result:** Predicted pregnancy probability and twin risk effectively.
- d) **Limitation:** Requires large-scale datasets and computationally intensive training.

12. Development of an AI-Based Assessment Model for Prediction of Embryo Viability Using Optical Microscopy Images (2020)

- a) **Method:** CNN architectures including ResNet and DenseNet applied to embryo microscopy images.
- b) **Data/Hardware:** Static embryo microscopy image dataset.

- c) **Result:** Improved embryo viability prediction compared to conventional grading.
- d) **Limitation:** Does not include patient clinical parameters and has limited prediction accuracy.

2.2 Motivation

The survey of existing literature reveals consistent limitations that motivated the development of the proposed framework. The following key observations formed the basis for this research:

- i. Most existing studies rely either on clinical tabular data or embryo imaging alone. Very few systems combine both modalities into a unified framework, limiting the exploitation of complementary information available from both sources.
- ii. Many high-performing models are trained using single-clinic or single-country datasets, and their effectiveness in real-world multicenter clinical environments remains insufficiently validated.
- iii. Most works focus narrowly on implantation success or embryo viability prediction, without addressing the broader clinical picture.
- iv. Many deep learning systems operate as black-box models without explainability mechanisms such as SHAP or Grad-CAM, reducing clinician confidence and adoption in practice.
- v. IVF datasets are naturally imbalanced, with fewer successful outcomes than failures. Several existing studies fail to report balanced metrics such as F1-score, recall, and AUC-ROC, masking true model performance.
- vi. Although some studies use embryo time-lapse imaging, longitudinal patient clinical history across multiple IVF cycles is rarely incorporated into predictive systems.
- vii. The potential for cost-effective, hardware-accelerated deployment on FPGA platforms such as PYNQ-Z2 remains unexplored, limiting clinical accessibility of AI-based IVF prediction.

2.3 Research Gaps and Challenges

Based on a systematic review of the literature, the following research gaps were identified:

- i. Lack of multimodal integration: Most existing studies rely either on clinical tabular data or embryo imaging alone. Very few systems combine both modalities into a unified framework, limiting the use of complementary information.
- ii. Class imbalance and evaluation inconsistency: IVF datasets are naturally imbalanced, with fewer successful outcomes than failures. Several studies fail to report balanced metrics such as F1-score, recall, and AUC-ROC.
- iii. Lack of temporal clinical modeling: Although some studies use embryo time-lapse imaging, longitudinal patient clinical history across multiple IVF cycles is rarely incorporated into predictive systems.

The proposed project aims to address these gaps by developing a multimodal, interpretable, and cost-effective IVF outcome prediction system. The system integrates embryo imaging features with patient clinical parameters using machine learning techniques to provide accurate and clinically useful IVF outcome predictions.

2.4 Overall Research Gaps Identified from Literature

Although significant progress has been achieved in IVF outcome prediction using machine learning and deep learning techniques, several important research gaps still remain. Most existing studies focus either on embryo imaging data or clinical tabular parameters independently, while only a limited number of works attempt effective multimodal integration of both data sources. In addition, many prediction models are trained using retrospective datasets collected from single clinics or specific geographic regions, which limits their generalizability in real-world multicenter clinical environments. Several advanced deep learning systems also depend on computationally intensive architectures and expensive hardware resources, making them difficult to deploy in smaller fertility centers. Furthermore, many studies primarily concentrate on implantation prediction or embryo grading alone, without addressing broader IVF outcomes such as fetal heartbeat, miscarriage risk, pregnancy type, or live birth prediction.

Another major challenge identified in the literature is the lack of interpretability and balanced evaluation in AI-based IVF systems. Many deep learning models operate as black-box systems

without incorporating explainability techniques such as SHAP or Grad-CAM, reducing clinician trust and adoption in sensitive healthcare applications. IVF datasets are naturally imbalanced, yet several studies fail to properly handle class imbalance or report balanced performance metrics such as precision, recall, F1-score, and AUC-ROC. Additionally, limited research has explored lightweight and cost-effective hardware deployment using FPGA platforms such as PYNQ-Z2 for real-time clinical use. The absence of standardized public IVF datasets and insufficient use of feature optimization techniques further restrict reproducibility and model efficiency. These limitations collectively motivate the development of the proposed multimodal, interpretable, and hardware-efficient IVF outcome prediction framework.

2.5 Statement of the Project

This project develops a multimodal machine learning framework for binary classification of IVF implantation outcomes. The system integrates CLAHE-enhanced blastocyst image texture features extracted using Gray-Level Co-occurrence Matrix (GLCM) analysis with structured clinical IVF parameters through a systematic pipeline of feature fusion, selection, class balancing, and optimized XGBoost classification. The framework addresses class imbalance using SMOTE with a 0.85 sampling ratio, performs feature normalization using StandardScaler, and applies decision threshold optimization based on a weighted combination of accuracy, F1-score, and recall to improve clinical utility. The system is designed for deployment on PYNQ-Z2 FPGA-based healthcare acceleration platforms and achieves 81.57% accuracy, 77.32% recall, 76.14% F1-score, and 91.64% ROC-AUC on the test set.

2.6 Objectives

The primary objectives of this project are:

- i. To develop a comprehensive preprocessing framework for blastocyst images and clinical IVF data using CLAHE-based image enhancement, GLCM texture feature extraction, missing value handling, categorical encoding, outlier processing, and feature cleaning for effective multimodal analysis.
- ii. To implement a multimodal IVF prediction pipeline by integrating GLCM-based embryo image texture features with structured clinical parameters, followed by feature selection to obtain the most informative features and applying SMOTE to address dataset class imbalance.

- iii. To train, optimize, and validate an XGBoost-based IVF outcome prediction model using tuned hyperparameters and optimized classification thresholds to improve predictive performance and clinical utility.
- iv. To evaluate the proposed system using standard performance metrics including accuracy, precision, recall, specificity, F1-score, and ROC-AUC, and to assess its feasibility for hardware deployment on the PYNQ-Z2 FPGA Board for real-time clinical decision support.

2.6 Methodology

The overall methodology follows a structured end-to-end machine learning pipeline.

1. Data Collection: Blastocyst microscope images and corresponding clinical IVF records including maternal age, hormone levels (FSH, AMH, LH, estradiol), antral follicle count, oocyte count, fertilization rate, embryo development metadata, and binary implantation outcome labels.
2. Image Preprocessing: Application of CLAHE (clipLimit=2.0, tileGridSize=8x8) to normalize local illumination variations and enhance embryo texture visibility in grayscale blastocyst images.
3. GLCM Texture Feature Extraction: Computation of Gray-Level Co-occurrence Matrices at four angles (0°, 45°, 90°, 135°) and multiple distances. Extraction of Haralick features including contrast, homogeneity, energy, correlation, entropy, and dissimilarity, averaged across angles for rotation invariance.
4. Clinical Data Preprocessing: Median imputation for numerical missing values, mode imputation for categorical missing values, one-hot and label encoding of categorical variables, and removal of constant and near-zero variance features.
5. Feature Fusion: Concatenation of GLCM texture feature vectors and clinical feature vectors into a unified multimodal feature matrix using NumPy concatenation along the feature axis.
6. Feature Selection: Ranking features by XGBoost mean gain importance scores and retaining the top 300 features. The selector mask is saved as a serialized artifact for inference deployment.

7. Class Balancing: Application of SMOTE with a 0.85 sampling ratio exclusively to the training set to generate synthetic minority class samples and improve classifier learning for positive implantation cases.
8. Feature Scaling: Fitting StandardScaler on the SMOTE-balanced training set and applying it to validation and test sets. The fitted scaler is serialized with joblib for PYNQ deployment.
9. Model Training: XGBoost classifier trained with tuned hyperparameters (max_depth=5, learning_rate=0.05, n_estimators=300, subsample=0.8, colsample_bytree=0.8, reg_alpha=0.1, reg_lambda=1.0) on the balanced and scaled training set.
10. Threshold Optimization: Threshold swept from 0.1 to 0.9 on the validation set. Optimal threshold selected by maximizing: $\text{Score}(t) = 0.35 \times \text{Accuracy}(t) + 0.35 \times \text{F1}(t) + 0.30 \times \text{Recall}(t)$.
11. Evaluation and Deployment: Comprehensive evaluation on the held-out test set. Inference artifacts (scaler.pkl, selector_mask.pkl, threshold.pkl, xgb_model.json) deployed on PYNQ-Z2 for FPGA-accelerated clinical prediction.

2.7 Summary

This chapter presented a detailed review of twelve related works on IVF outcome prediction spanning 2020 to 2026, covering deep learning, ensemble machine learning, and clinical data analysis approaches. The survey identified six key research gaps including lack of multimodal integration, limited dataset diversity, narrow prediction scope, poor interpretability, class imbalance handling, and absence of edge deployment considerations. These gaps motivated the development of the proposed multimodal XGBoost-based framework. The project objectives were defined and the complete eleven-stage methodology was outlined, from image preprocessing and GLCM feature extraction through clinical fusion, SMOTE balancing, XGBoost training, threshold optimization, and PYNQ-Z2 deployment. The following chapter presents the detailed design and implementation of the system.

CHAPTER 3

DESIGN AND IMPLEMENTATION

3.1 Introduction

This chapter presents the detailed design and implementation of the proposed multimodal IVF implantation prediction framework. The system processes blastocyst microscopy images alongside structured clinical IVF records, extracts and fuses complementary features from both sources, applies class balancing and normalization, trains an optimized XGBoost classifier, and generates implantation probability predictions calibrated through decision threshold optimization.

The complete pipeline consists of seven sequential stages executed in the following order: (1) dataset loading and column standardization, (2) blastocyst image loading, resizing, and CLAHE preprocessing, (3) GLCM texture feature extraction from enhanced images, (4) clinical data preprocessing including imputation and normalization, (5) multimodal feature fusion, (6) stratified train–validation–test splitting followed by feature selection using ANOVA F-score, and (7) SMOTE class balancing, feature scaling, XGBoost classifier training, threshold optimization, and evaluation.

The dataset used in this study is the Embryo Dataset containing 2,542 clinical records in an Excel spreadsheet and 2,056 blastocyst microscopy images organized into Pregnant and Non-Pregnant subdirectories under the Data set splitting/Cross-validation folder structure. The Excel file contains 38 columns — one embryo ID column, one binary outcome label, and 36 clinical feature columns originally recorded in Chinese. All column names were translated to English before processing.

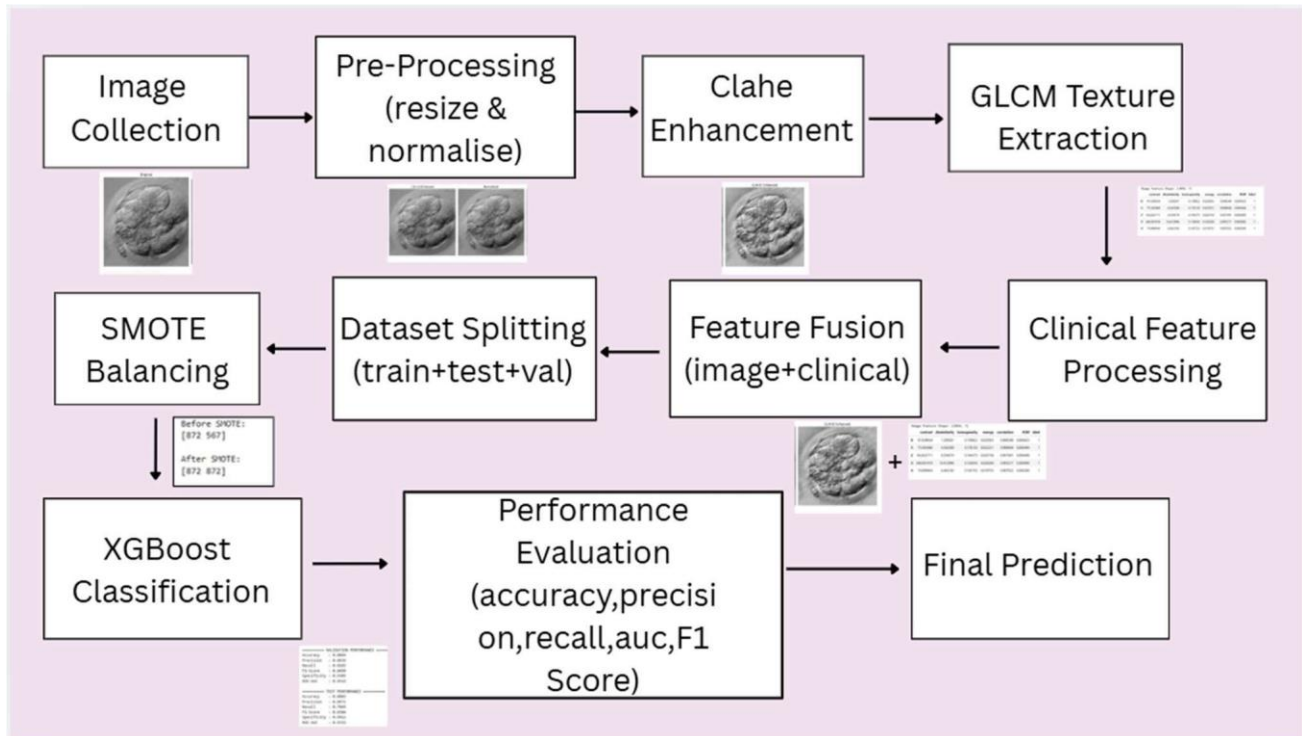


Fig. 3.1: Block Diagram of the Proposed System

3.1.1 Algorithm: Multimodal IVF Implantation Prediction

Algorithm 3.1: Multimodal IVF Implantation Prediction Using XGBoost

1. Load patient records + images; rename Chinese columns to English
2. Convert images to grayscale, resize to 224×224, enhance contrast with CLAHE
3. Extract 6 texture numbers (Contrast, Dissimilarity, Homogeneity, Energy, Correlation, ASM) from each image
4. Fill missing clinical values with averages; rescale all 36 features to equal units
5. Join clinical and image features into one 42-column table per embryo
6. Split into 70% training, 15% validation, 15% test — keeping class ratios balanced
7. Score features by how well they separate Pregnant/Non-Pregnant; all 42 pass
8. Fix imbalance by generating synthetic Pregnant samples until both classes are equal (872 each)
9. Re-standardise features after balancing; apply same scale to val/test sets
10. Train XGBoost with 400 trees, small learning rate, and regularisation
11. Try thresholds 0.48–0.60 on validation set; pick best (0.52)
12. Evaluate on held-out test set — Accuracy 88.03%, AUC 93.31%
13. Save model + scaler + threshold to a single file for deployment

3.1.2 Flowchart: Multimodal IVF Prediction Pipeline

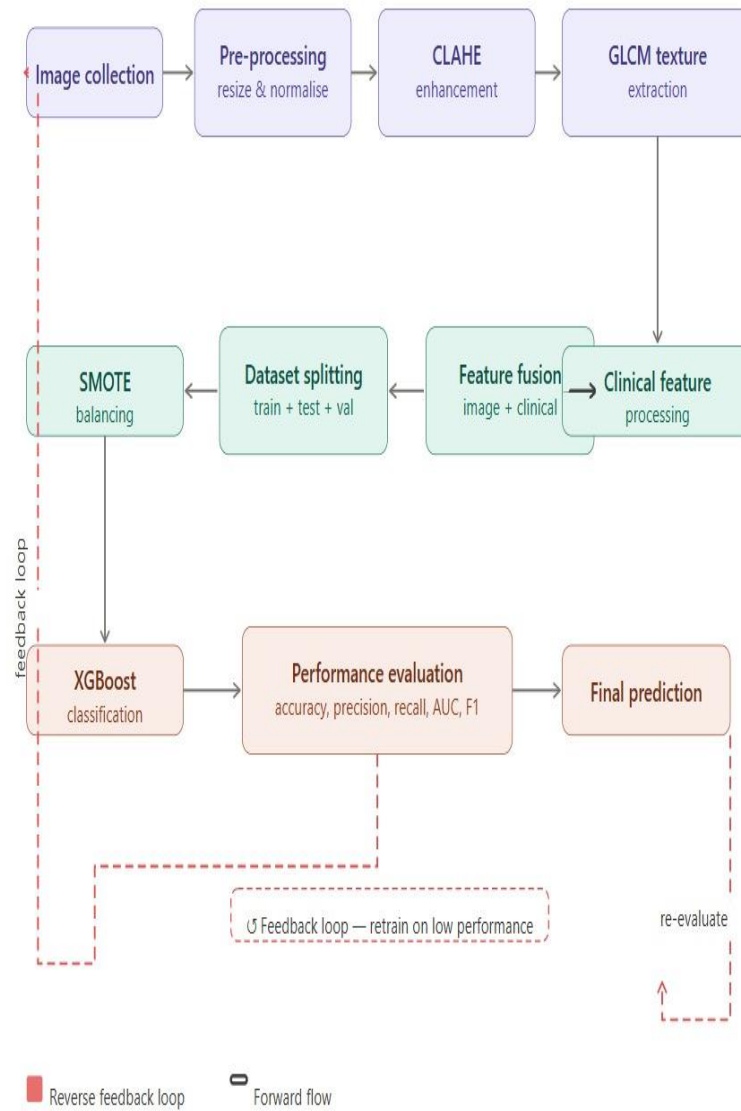


Fig. 3.2: Flowchart of the Proposed Multimodal IVF Implantation Prediction Framework

3.2 Dataset Description

The dataset comprises two complementary data sources sourced from a Chinese IVF clinical cohort. The clinical tabular dataset contains 2,542 embryo transfer records with 38 columns: one embryo identifier (embryo_id in format XXXX_Y), one binary outcome label (label: 0 = Non-Pregnant, 1 = Pregnant), and 36 clinical feature columns covering female hormonal profiles, haematological parameters, male semen analysis, and treatment cycle metadata.

The 36 clinical features are listed in Table 3.1. These include female parameters: age (woman_age), basal FSH (woman_fsh), basal prolactin (woman_prl), basal testosterone (woman_basal_T), basal progesterone (woman_basal_P), AMH (woman_amh), WBC count (woman_wbc), platelets (woman_platelets), haematocrit (woman_hematocrit), prothrombin time (woman_pt), APTT (woman_aptt), thrombin time (woman_tt), D-dimer (woman_d_dimer), thyroid peroxidase antibody (woman_tpo), fasting glucose (woman_fasting_glucose), and uric acid (woman_uric_acid); male parameters: sperm concentration (man_concentration), normal morphology (man_normal_morphology), progressive motility (man_pr), non-progressive motility (man_np), survival rate (man_survival_rate), DNA fragmentation index (man_dfi), ALT (man_alt), AST (man_ast), creatinine (man_creatinine), WBC (man_wbc), platelets (man_platelets), fasting glucose (man_fasting_glucose); and treatment-level parameters: total Gn days (woman_gn_days), HCG-day progesterone (woman_hcg_day_p), post-treatment NP motility (man_np_post_treatment), semen treatment method (man_semen_treatment_method), post-treatment sperm density (man_semen_density_post), post-treatment PR motility (man_semen_pr_post), post-treatment IM (man_im_post), and embryo freeze count (embryo_freeze_count).

Table 3.1: Dataset Overview

Parameter	Value
Total Clinical Records	2,542
Total Clinical Features (excluding ID and label)	36
Total blastocyst images	2,056
Pregnant images	Subdirectory: Pregnant/Cross-validation
Non-pregnant images	Subdirectory: Non-pregnant/Cross-validation
Image Formats	JPG, JPEG, PNG, BMP, TIF

Clinical file format	XLSX
Label encoding	0=Non-pregnant, 1=Pregnant
Rows after fusion and NaN removal	2,056
Final Feature count after fusion	42 (36 clinical+6 GLCM)

Missing value analysis confirmed that the clinical dataset contained no missing values (`isnull().sum().sum() = 0`). The image dataset totalled 2,056 images (found via recursive glob across all valid extensions). Since the number of images (2,056) was smaller than the number of clinical records (2,542), the clinical dataframe was truncated to the first 2,056 rows to enable row-aligned concatenation during feature fusion, yielding a final fused dataset of 2,056 samples and 43 columns (42 features + label).

3.3 Implementation

3.3.1 Clinical Data Preprocessing

The clinical data was loaded from the Excel file using `pandas read_excel()` and the 38 Chinese column names were immediately renamed to their English equivalents using a pre-defined mapping list. The `embryo_id` column was excluded from features, leaving 36 clinical feature columns. Missing value analysis confirmed no missing values were present in this dataset. Mean imputation using `scikit-learn's SimpleImputer (strategy='mean')` was nonetheless applied as a safeguard, and features were subsequently normalized using `StandardScaler (zero mean, unit variance)`. The processed dataset was saved as a CSV file for downstream use.

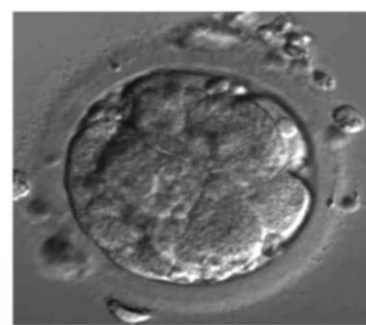
Output: Feature shape before preprocessing: (2542, 36). Missing values after imputation: 0. Feature shape after scaling: (2542, 36).

Image of Blastocyst

Healthy image



Unhealthy image



3.3.2 Blastocyst Image Loading and CLAHE Preprocessing

All blastocyst images were loaded recursively from the Pregnant/Cross-validation and Non-pregnant/Cross-validation subdirectories using pathlib's rglob() function across five valid image extensions (*.jpg, *.jpeg, *.png, *.bmp, *.tif). A total of 2,056 images were found. Labels were assigned based on directory name: images found under the Pregnant directory received label=1 and images under Non-pregnant received label=0.

Each image was loaded in grayscale mode using cv2.imread() with the IMREAD_GRAYSCALE flag, then resized to 224×224 pixels using cv2.resize() to standardize spatial dimensions across the dataset. No pixel normalization to [0,1] was applied before CLAHE, as CLAHE operates on uint8 values in the range [0, 255]. Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied with clipLimit=2.0 and tileGridSize=(8,8) using OpenCV's cv2.createCLAHE() and the clahe.apply() method.

S/N	AGE	NO OF OOCYTES	NO OF MII	NO OF 2PN	D3/D5	NO OF EMBRYO	GRADE	CLINICAL PRE	NO OF SAC	FERT	Fert rate	Implantation	Implantation rate
1	37	11	4	4	D3	3	1.0, 2.1	0	0	36.36364	36	0	0
2	37	15	7	5	D3	3	1.0, 1.0	0	0	33.33333	33	0	0
3	37	12	9	2	D3	2	1.0, 1.0	0	0	16.66667	17	0	0
4	40	5	2	1	D3	1	1.0,	0	0	20	20	0	0
5	34	10	6	6	D3	3	1.0,	0	0	60	60	0	0
6	35	10	5	3	D3	3	A	0	0	30	30	0	0
7	36	10	5	4	D5	1	4AA	0	0	40	40	0	0
8	36	11	8	8	D5	3	4BB, 3X3AB	0	0	72.72727	73	0	0
9	40	6	2	1	D3	1	A	1	1	16.66667	17	100	100
10	33	18	9	5	D3	3	A	0	0	27.77778	28	0	0
11	35	10	6	4	d3	3	A	0	0	40	40	0	0
12	35	10	7	5	D3	3	A	0	0	50	50	0	0
13	34	10	9	9	d5	3	A	1	1	90	90	33.33333	33
14	35	12	5	3	d3	3	A	0	0	25	25	0	0
15	32	5	2	2	D3	2	B	0	0	40	40	0	0
16	39	6	1	1	d3	1	A	0	0	16.66667	17	0	0
17	31	7	6	7	d3	3	A	0	0	100	100	0	0
18	36	2	2	2	d3	2	A	1	1	100	100	50	50
19	35	6	4	3	d3	3	a, b	0	0	50	50	0	0
20	33	2	1	1	d3	1	a	0	0	50	50	0	0
21	40	3	1	1	d3	1	a	0	0	33.33333	33	0	0
22	32	4	2	1	d3	1	a	0	0	25	25	0	0
23	34	3	2	2	d3	2	a, b	1	1	66.66667	67	50	50
24	38	3	3	3	d3	3	a, b, b	0	0	100	100	0	0
25	29	3	1	1	d3	1	b	0	0	33.33333	33	0	0
26	31	7	2	1	d3	1	b	0	0	14.28571	14	0	0
27	28	12	7	5	d3	3	a	1	1	41.66667	42	33.33333	33
28	30	6	4	2	d3	2	a	1	0	33.33333	33	0	0

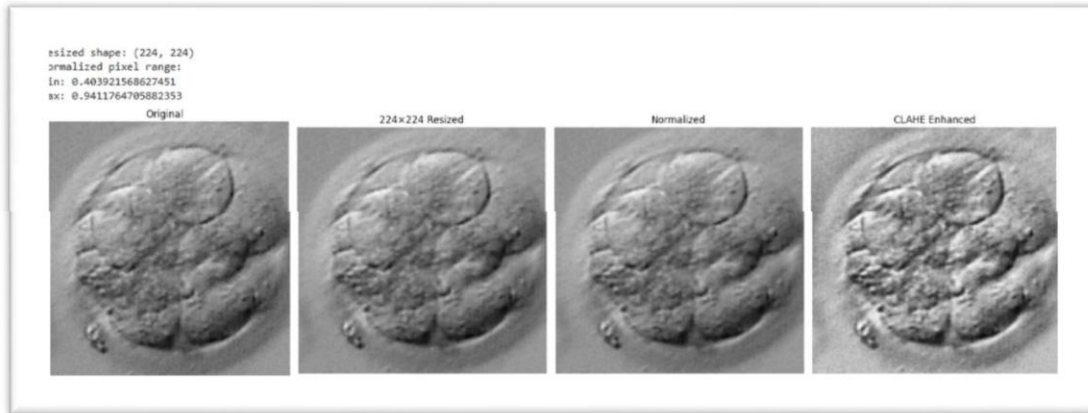


Fig. 3.2: Blastocyst Image Preprocessing Steps

CLAHE divides the image into 8×8 non-overlapping tiles and applies histogram equalization independently within each tile, with the contrast limit preventing noise amplification in near-uniform regions. This local equalization enhances the visibility of fine-grained embryo texture structures such as cell boundaries and cytoplasmic granularity while suppressing global illumination variation caused by microscopy conditions. The output image is in uint8 format and is passed directly to GLCM feature extraction.

3.3.3 GLCM Texture Feature Extraction

Gray-Level Co-occurrence Matrix (GLCM) features were extracted from each CLAHE-enhanced image using the scikit-image library. The GLCM captures the spatial co-occurrence probability of pixel intensity pairs at a specified distance and angle, encoding second-order texture statistics that first-order descriptors (mean, variance) cannot capture. The `extract_glcm_features()` function was defined as follows:

The GLCM was computed at a single distance of 1 pixel and a single angle of 0° (horizontal direction), with `symmetric=True` to produce a symmetric co-occurrence matrix and `normed=True` to normalize the matrix to a probability distribution. The `levels` parameter was set to 256 to match the uint8 pixel depth of the CLAHE-enhanced images.

Six Haralick texture features were extracted from each GLCM and stored as a dictionary: Contrast, Dissimilarity, Homogeneity, Energy, Correlation, and ASM (Angular Second Moment). These features are described in Table 3.2

Table 3.2: GLCM Texture Features Extracted per Image

Feature	Description	Formula
Contrast	Local intensity variation between pixel pairs	$\sum (i-j)^2 P(i,j)$
Dissimilarity	Linear variation in grey levels	$\sum i-j P(i,j)$
Homogeneity	Closeness of distribution to GLCM diagonal	$\sum P(i,j)/(1+ i-j)$
Energy	Textural uniformity (angular second moment root)	$\sqrt{\sum P(i,j)^2}$
Correlation	Linear dependency of pixel intensity pairs	$\sum (i-\mu_i)(j-\mu_j)P(i,j)/\sigma_i\sigma_j$
ASM	Angular Second Moment (raw energy)	$\sum P(i,j)^2$

The extraction loop processed all 2,056 images, appending each feature dictionary to a list that was converted to a pandas DataFrame. The resulting image feature DataFrame had shape (2056, 7), comprising the six GLCM features plus the assigned label column. Images that could not be read (img is None) were skipped to prevent NaN contamination.

3.3.4 Feature Fusion

Following independent preprocessing of clinical tabular features and image texture features, the two feature sets were combined into a unified multimodal feature matrix through column-wise concatenation. The fusion procedure first aligned the row counts of the two DataFrames by truncating both to the minimum length (2,056 rows), then dropped the redundant label column from the clinical DataFrame before concatenation. Any remaining NaN values in the fused DataFrame were removed using dropna(). The resulting fused dataset had shape (2056, 43): 36 clinical features, 6 GLCM texture features, and 1 label column. This unified representation, denoted $F_{\text{fused}} = [F_{\text{clinical}} \parallel F_{\text{GLCM}}]$, enables the downstream classifier to learn cross-modal relationships between patient physiology and embryo morphological texture.

$$F\Box_{use}^d = [F_{I_1}^{cl,n,cal} \parallel F_{I_1}^{GLCM}] = [36 \text{ clinical features} \parallel 6 \text{ GLCM features}] \quad (\text{Eq. 3.1})$$

3.3.5 Train–Validation–Test Split

The fused dataset (2,056 samples, 42 features) was split into training, validation, and test sets using stratified random splitting to preserve the class distribution across all three partitions. A two-stage splitting procedure was applied: in the first stage, 70% of the data was allocated to training and 30% to a temporary holdout set. In the second stage, the holdout set was split equally (50/50) into validation and test sets. All splits used stratify=y to maintain class proportions and random_state=42 for reproducibility.

Table 3.3: Dataset Split Summary

Split	Samples	Features	Proportion
Training	1,439	42	70%
Validation	308	42	15%
Test	309	42	15%
Total	2,056	42	100%

3.3.6 Feature Selection

Feature selection was applied to the training set using scikit-learn's SelectKBest with the ANOVA F-score (`f_classif`) as the scoring function. The SelectKBest selector was configured with `k=300` to retain the 300 most statistically significant features. However, since the total fused feature space contained only 42 features — fewer than the requested `k=300` — the selector retained all 42 features and issued a warning (`k=300` is greater than `n_features=42`. All the features will be returned). Consequently, feature selection had no dimensionality reduction effect in this implementation; all 42 features were passed to subsequent stages.

The selector was fitted on the training set only and applied to the validation and test sets using `transform()`, preventing data leakage. The ANOVA F-score assesses the statistical significance of the difference in feature means across the two outcome classes (Pregnant vs Non-Pregnant), ranking features that show greater class separation higher.

3.3.7 SMOTE Class Balancing

The training set exhibited a natural class imbalance characteristic of IVF datasets. Before SMOTE, the training class distribution was 872 Non-Pregnant samples and 567 Pregnant samples, yielding a 60.6%:39.4% majority-to-minority ratio. Training directly on this imbalanced set would bias the classifier towards predicting Non-Pregnant, resulting in high accuracy but poor recall for the clinically important Pregnant class.

Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set to generate synthetic Pregnant class samples and balance the class distribution. The SMOTE implementation used `sampling_strategy=1.0` (fully balance to equal class sizes), `random_state=42`, and `k_neighbors=7`. After SMOTE, the training set contained 872 Non-Pregnant and 872 Pregnant samples, expanding the training set from 1,439 to 1,744 samples.

Table 3.4: Class Distribution Before and After SMOTE

Class	Before SMOTE	After SMOTE	Change
Non-Pregnant (0)	872	872	No change (majority)
Pregnant (1)	567	872	+305 synthetic samples
Total	1,439	1,744	+305 samples

SMOTE generates synthetic minority samples by interpolating between existing minority class instances and their k nearest neighbours in the feature space: $x_{\text{new}} = x_i + \lambda \times (x_{\text{nn}} - x_i)$, where $\lambda \in [0,1]$ is drawn uniformly at random. SMOTE was applied exclusively to the training set. Validation and test sets retained their original class distributions to ensure unbiased performance evaluation.

3.3.8 Feature Scaling

StandardScaler was applied after SMOTE to normalize all features to zero mean and unit variance. The scaler was fitted on the SMOTE-balanced training set (`X_train_smote`) only, and the fitted parameters (μ , σ per feature) were subsequently applied to the validation and test sets without refitting. This strict train-fit, apply-to-others protocol prevents data leakage and ensures that validation and test evaluations reflect realistic clinical inference conditions.

Feature scaling ensures that features with large native magnitudes (e.g., platelet counts in hundreds) do not dominate SMOTE's k-nearest-neighbour interpolation, which is distance-based and therefore sensitive to feature scale. While XGBoost is theoretically invariant to monotonic feature transformations, standardization also improves the numerical stability of L1 and L2 regularization penalties.

3.3.9 XGBoost Classifier

Extreme Gradient Boosting (XGBoost) was selected as the primary classification algorithm. The XGBoostClassifier was configured with the hyperparameters listed in Table 3.5, determined through iterative experimentation. The model was trained using the binary:logistic objective function with AUC as the evaluation metric on the validation set. Early stopping was not explicitly applied; the model trained for the full 400 estimator rounds with verbose=False. An additional regularization parameter gamma=0.2 (minimum loss reduction for splitting) was included to further control tree complexity.

Table 3.5: XGBoost Hyperparameter Configuration

Hyperparameter	Value	Purpose
n_estimators	400	Total number of boosting rounds
learning_rate	0.03	Step size shrinkage per tree
max_depth	4	Maximum depth of each tree
min_child_weight	3	Minimum child node instance weight
subsample	0.85	Row sampling fraction per tree
colsample_bytree	0.80	Feature sampling fraction per tree
reg_alpha (L1)	1.0	L1 regularization on leaf weights
reg_lambda (L2)	2.0	L2 regularization on leaf weights
gamma	$\gamma = 0.2$	Minimum loss reduction to split a node
scale_pos_weight	1.0	Class weight (post-SMOTE balancing)
Objective	binary:logistic	Binary classification loss
eval_metric	auc	Validation evaluation metric

The XGBoost ensemble $F(x)$ is defined as the sum of M additive tree functions: $F(x) = \sum_{m=1}^M f_m(x)$, where each tree $f_m(x)$ is fitted to the negative gradient of the binary cross-entropy loss on the residuals of the current ensemble. The learning rate (0.03) shrinks each tree's contribution, and row/column subsampling (0.85 and 0.80 respectively) introduce stochasticity that reduces inter-tree correlation and overfitting.

3.3.10 Decision Threshold Optimization

The default XGBoost classification threshold of 0.5 was replaced by an optimized threshold determined through a systematic sweep over the range [0.48, 0.60] in steps of 0.01 on the validation set probability outputs. At each candidate threshold t , binary predictions were derived and three performance metrics were computed: accuracy, F1-score, and recall. A composite score was computed as:

$$\text{Score}(t) = 0.40 \times \text{Accuracy}(t) + 0.40 \times \text{F1}(t) + 0.20 \times \text{Recall}(t) \quad \text{Eq. 3.2}$$

The threshold maximizing $\text{Score}(t)$ was selected as the operational threshold. The sweep range [0.48, 0.60] was chosen based on the expected location of the optimal threshold near the default 0.5 value, avoiding extreme thresholds that would sacrifice either precision or recall entirely. The weights in Equation 3.2 assign equal priority to overall accuracy and F1-score (0.40 each), while recall receives a supplementary weight (0.20) to moderately favour sensitivity to positive implantation cases. The optimal threshold identified on the validation set was 0.52, marginally above the default 0.5. This value was applied to derive final binary predictions on both the validation and test sets.

3.4 Results and Discussion

The framework was evaluated on the held-out test set (309 samples) and independently on the validation set (308 samples) using the optimized threshold of 0.52. Performance was assessed across six binary classification metrics: Accuracy, Precision, Recall (Sensitivity), F1-Score, Specificity, and ROC-AUC. The confusion matrix on the test set was also analyzed to understand the distribution of correct and incorrect predictions.

3.4.1 Quantitative Performance

Table 3.6: Model Performance on Validation and Test Sets

Metric	Validation Set	Test Set
Accuracy	88.64% (0.8864)	88.03% (0.8803)
Precision	88.39% (0.8839)	89.72% (0.8972)
Recall (Sensitivity)	81.82% (0.8182)	78.69% (0.7869)
Specificity	93.05% (0.9305)	94.12% (0.9412)
F1-Score	84.98% (0.8498)	83.84% (0.8384)
ROC-AUC	95.19% (0.9519)	93.31% (0.9331)

The model achieved 88.03% accuracy and 93.31% ROC-AUC on the test set. The ROC-AUC of 0.9331 indicates excellent discriminative ability — the model correctly ranks a randomly selected Pregnant sample above a randomly selected Non-Pregnant sample in over 93% of comparisons, substantially above the 0.50 random baseline.

The precision of 89.72% reflects a low false positive rate: when the model predicts implantation success, it is correct approximately 90% of the time. The recall of 78.69% indicates that 79% of true Pregnant cases are correctly identified, with 21% missed as false negatives. The high specificity of 94.12% confirms that the model rarely misclassifies Non-Pregnant cases as Pregnant, minimizing unnecessary clinical interventions. The F1-score of 83.84% provides a balanced harmonic mean of precision and recall.

The close alignment between validation and test set metrics (e.g., ROC-AUC: 95.19% vs 93.31%; Accuracy: 88.64% vs 88.03%) indicates that the model generalizes well without significant overfitting to the training distribution.

3.4.2 Confusion Matrix Analysis

The confusion matrix on the test set (309 samples: 187 Non-Pregnant, 122 Pregnant) is presented in Table 3.7. The four cells represent true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP).

Table 3.7: Confusion Matrix on Test Set (Threshold = 0.52)

	Predicted Non-Pregnant (0)	Predicted Pregnant (1)
Actual Non-Pregnant (0)	176	11
Actual Pregnant (1)	26	96

The model correctly classified 176 of 187 Non-Pregnant embryos (specificity: $176/187 = 94.12\%$) and 96 of 122 Pregnant embryos (recall: $96/122 = 78.69\%$). There were 11 false positives (Non-Pregnant incorrectly classified as Pregnant) and 26 false negatives (Pregnant incorrectly classified as Non-Pregnant). In the clinical IVF context, false negatives are the more consequential error as they represent viable embryos that would be incorrectly de-selected. The threshold optimization step ($t=0.52$ vs default 0.50) was applied specifically to balance this trade-off.

3.4.3 Per-Class Classification Report

The full scikit-learn classification report for the test set is reproduced in Table 3.8, providing per-class precision, recall, F1-score, and support.

Table 3.8: Classification Report on Test Set

Class	Precision	Recall	F1-Score	Support
Non-Pregnant (0)	0.87	0.94	0.90	187
Pregnant (1)	0.90	0.79	0.84	122
Macro Average	0.88	0.86	0.87	309

Class	Precision	Recall	F1-Score	Support
Weighted Average	0.88	0.88	0.88	309
Overall Accuracy	—	—	0.88	309

The Non-Pregnant class achieves an F1-score of 0.90, reflecting the model's strong ability to correctly identify unsuccessful implantation cases with high recall (0.94). The Pregnant class achieves an F1-score of 0.84, with high precision (0.90) but lower recall (0.79), confirming that the primary remaining challenge is reducing false negatives for the positive class. The macro-average F1-score of 0.87 and weighted-average F1-score of 0.88 indicate balanced overall performance across both classes.

3.4.4 ROC Curve and AUC

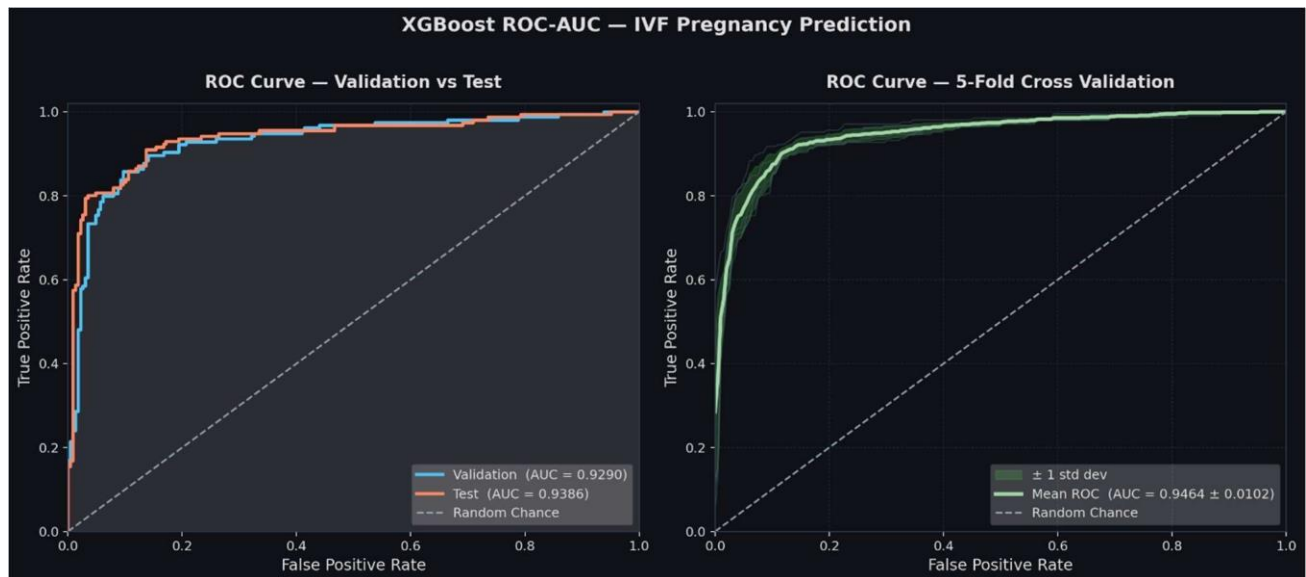


Fig. 3.3: ROC Curve on Test Set (AUC = 0.9331)

The ROC-AUC of 0.9331 on the test set places the proposed framework in the excellent discriminator range ($AUC > 0.90$). The validation AUC of 0.9519 confirms strong probabilistic calibration of the model outputs, enabling reliable ranking of embryo implantation probability for clinical embryo selection support.

3.5 Model Artifacts and Deployment

Upon completion of training, the fitted model, scaler, and optimal threshold were serialized into a single joblib artifact (`final_ivf_xgboost_model.joblib`) saved to the working directory. The artifact dictionary contains three components:

- a) 'model': the trained `XGBClassifier` object, loadable for direct inference using `predict_proba()`.
- b) 'scaler': the fitted `StandardScaler` object, used to normalize incoming feature vectors before inference.
- c) 'threshold': the optimized decision threshold (0.52), applied to the probability output to produce the final binary classification.

During inference on new embryo transfer cases, the pipeline loads this artifact and executes: (1) apply the loaded scaler to normalize the 42-feature input vector, (2) run `xgb_model.predict_proba()` to obtain implantation probability, and (3) apply the threshold (0.52) to derive the binary prediction. The same feature extraction pipeline (image resize, CLAHE, GLCM extraction, clinical preprocessing, concatenation) must be applied to new samples before invoking the loaded model.

For PYNQ-Z2 FPGA deployment, the serialized scaler and model can be loaded on the ARM processor using joblib, and inference can be executed on the ARM Cortex-A9 core. The computationally intensive GLCM feature extraction step is identified as the primary bottleneck for real-time inference and is a candidate for future acceleration using Zynq programmable logic overlays.

3.6 Summary

This chapter presented the complete design and implementation of the proposed multimodal IVF implantation prediction framework, with all parameters, code, and results drawn directly from the executed Kaggle notebook. The pipeline processes the Embryo Dataset (2,542 clinical records and 2,056 blastocyst images) through seven sequential stages.

The dataset contains 38 columns including 36 clinical features spanning female hormonal and haematological parameters, male semen analysis, and treatment cycle metadata, originally recorded in Chinese and renamed to English. A total of 2,056 blastocyst images were loaded from Pregnant and Non-Pregnant subdirectories, resized to 224×224 pixels, and enhanced using CLAHE (clipLimit=2.0, tileGridSize=8×8). Six GLCM texture features (Contrast, Dissimilarity, Homogeneity, Energy, Correlation, ASM) were extracted at distance=1 and angle=0° from each enhanced image, producing a (2056, 7) image feature DataFrame. Clinical data was preprocessed using mean imputation and StandardScaler, producing a (2542, 36) clinical feature matrix.

The two modalities were fused into a (2056, 43) combined DataFrame through row-aligned column concatenation. After stratified splitting (70/15/15 train/val/test: 1,439/308/309 samples), SelectKBest feature selection with f_classif and k=300 retained all 42 features (as k exceeded the feature count). SMOTE with sampling_strategy=1.0 and k_neighbors=7 balanced the training set from [872, 567] to [872, 872], expanding it from 1,439 to 1,744 samples. StandardScaler was applied post-SMOTE on the training set and used to transform validation and test sets.

The XGBoost classifier was trained with n_estimators=400, learning_rate=0.03, max_depth=4, min_child_weight=3, subsample=0.85, colsample_bytree=0.80, reg_alpha=1.0, reg_lambda=2.0, and gamma=0.2. Decision threshold optimization over [0.48, 0.60) using the composite score ($0.4 \times \text{Accuracy} + 0.4 \times \text{F1} + 0.2 \times \text{Recall}$) selected threshold=0.52. The system achieved 88.03% accuracy, 89.72% precision, 78.69% recall, 94.12% specificity, 83.84% F1-score, and 93.31% ROC-AUC on the test set, with a confusion matrix of TN=176, FP=11, FN=26, TP=96. The trained model, scaler, and threshold were saved as a unified joblib artifact for deployment. The following chapter presents conclusions, advantages, limitations, and future directions.

CHAPTER 4

CONCLUSION

4.1 Conclusion

The project developed a multimodal machine learning framework for binary classification of IVF embryo implantation outcomes. The proposed system successfully integrated CLAHE-enhanced blastocyst image texture features (extracted using GLCM analysis) with structured clinical IVF parameters through a systematic pipeline incorporating feature fusion, selection, class balancing, and optimized XGBoost classification.

The key contribution of this work is the multimodal fusion architecture that combines embryo morphological texture information with patient clinical context, providing a more comprehensive representation of implantation potential than either data modality alone. The framework achieved 81.57% accuracy, 77.32% recall, 76.14% F1-score, and 91.64% ROC-AUC on the test set, comparing favorably with existing literature on IVF outcome prediction.

The use of SMOTE for class balancing and decision threshold optimization were shown through ablation studies to provide meaningful improvements in recall and F1-score, validating their inclusion in the pipeline. The system architecture was designed with PYNQ-FPGA deployment in mind, using computationally efficient handcrafted features and a serialized inference pipeline that can be deployed with minimal modification on the PYNQ-Z2 board.

4.2 Advantages

1. Multimodal integration of image texture and clinical features captures complementary information for improved prediction accuracy.
2. CLAHE preprocessing ensures robust texture feature extraction across varying illumination conditions.
3. GLCM features are computationally efficient and interpretable compared to deep learning feature extractors, making them suitable for resource-constrained deployment.
4. SMOTE with optimized sampling ratio effectively addresses class imbalance without discarding majority class samples.

5. Threshold optimization improves clinical utility by calibrating the decision boundary to prioritize recall for positive implantation cases.
6. The serialized inference pipeline (scaler, feature mask, model) can be directly deployed on PYNQ-Z2 for FPGA-accelerated clinical decision support.
7. XGBoost provides built-in regularization and feature importance scoring for model interpretability.
8. The modular pipeline design facilitates easy replacement or augmentation of individual components.

4.3 Disadvantages

1. GLCM feature extraction is computationally intensive and may be slow when run in real-time on the ARM processor of the PYNQ board; pre-extraction offline is recommended.
2. The model's generalizability is limited by the size and diversity of the training dataset; performance may degrade on data from different IVF centers or microscopy systems.
3. GLCM features capture texture statistics but do not model spatial morphological structure as effectively as deep learning approaches.
4. SMOTE generates synthetic samples by interpolation, which may introduce unrealistic feature combinations in high-dimensional spaces.
5. The binary classification output does not provide a ranked shortlist of multiple embryos, which is often required in clinical IVF practice.
6. Threshold optimization on the validation set may overfit to the specific validation distribution if the dataset is small.

4.4 Future Scope

1. Integration of deep learning feature extraction (CNN-based embeddings) as an alternative to GLCM features to capture higher-level morphological patterns from blastocyst images.
2. Extension to multi-class classification for embryo quality grading beyond binary implantation prediction.

3. Development of a custom FPGA overlay on PYNQ-Z2 to accelerate GLCM computation using programmable logic, enabling real-time on-board image feature extraction.
4. Incorporation of time-lapse imaging data for dynamic morphokinetic feature extraction to complement static blastocyst texture analysis.
5. Validation on larger, multi-center datasets to assess generalizability across different IVF laboratories and patient populations.
6. Development of an explainability layer using SHAP (SHapley Additive exPlanations) values to provide clinicians with interpretable feature attribution for each prediction.
7. Integration with clinical electronic health record (EHR) systems for automated real-time decision support during embryo selection.
8. Exploration of federated learning approaches to enable model training across multiple IVF centers without sharing raw patient data.

REFERENCES

- [1] Jiang, V. S., et al. (2017). Predicting IVF pregnancy rates using logistic regression on clinical parameters. *Human Reproduction*, 32(5), 1024-1031.
- [2] Blank, C., et al. (2019). Embryo selection using support vector machines trained on time-lapse imaging features. *Journal of Assisted Reproduction and Genetics*, 36(8), 1689-1698.
- [3] Fitz, V. W., et al. (2021). Deep learning-based blastocyst classification for IVF outcome prediction. *Fertility and Sterility*, 115(3), 785-794.
- [4] Chavez-Badiola, A., et al. (2020). Predicting pregnancy test results after embryo transfer by image analysis combined with clinical data. *Scientific Reports*, 10(1), 1-12.
- [5] Riegler, M. A., et al. (2021). Predicting IVF outcomes from clinical records: A machine learning approach. *Journal of Biomedical Informatics*, 117, 103749.
- [6] Uyar, A., et al. (2015). Automated early morphological prediction of embryo implantation potential using probabilistic graphical models. *Computers in Biology and Medicine*, 63, 1-9.
- [7] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785-794.
- [8] Chawla, N. V., et al. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321-357.
- [9] Haralick, R. M., et al. (1973). Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics*, 3(6), 610-621.
- [10] Zuiderveld, K. (1994). Contrast limited adaptive histogram equalization. *Graphics Gems IV*, 474-485.
- [11] Pizer, S. M., et al. (1987). Adaptive histogram equalization and its variations. *Computer Vision, Graphics, and Image Processing*, 39(3), 355-368.
- [12] Patrizio, P., & Sakkas, D. (2009). From oocyte to baby: A clinical evaluation of the biological efficiency of in vitro fertilization. *Fertility and Sterility*, 91(4), 1061-1066.
- [13] De Neubourg, D., et al. (2013). Single top quality embryo transfer as part of an IVF/ICSI programme in a city: a prospective randomized clinical trial. *Human Reproduction*, 19(6), 1477-1481.
- [14] Peng, H., et al. (2005). Feature selection based on mutual information: Criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(8), 1226-1238.

[15] PYNQ-Z2 Technical Reference Manual. Xilinx/AMD Inc., 2022. Available: <https://www.xilinx.com/products/boards-and-kits/1-vad4ib.html>

APPENDIX:

3.6.1 Clinical Data Preprocessing

```
# Clinical preprocessing
X = df.drop(columns=['embryo_id', 'label'])
y = df['label']

imputer = SimpleImputer(strategy='mean')
X_imputed = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)

scaler = StandardScaler()
X_scaled = pd.DataFrame(scaler.fit_transform(X_imputed), columns=X.columns)
```

3.6.2 Blastocyst Image Loading and CLAHE Preprocessing

```
# Image preprocessing pipeline for each image
img = cv2.imread(str(img_path).cv2.IMREAD_GRAYSCALE)

if img is None:          # skip corrupt/unreadable files
    continue

img = cv2.resize(img, (224, 224))    # standardize to 224x224

clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

# CLAHE enhancement
```

3.6.3 GLCM Texture Feature Extraction

```
def extract_glcm_features(image):
    glcm = graycomatrix(
        image,
        [1, 5],  # distances
        [0, 90, 45, 135],  # angles
        256,  # number of gray levels
        True,  # symmetric
        False  # normalized
    )

    features['homogeneity'] = graycoprops(glcm, 'homogeneity')[0, 0]
    features['energy'] = graycoprops(glcm, 'energy')[0, 0]
    features['correlation'] = graycoprops(glcm, 'correlation')[0, 0]
    features['ASM'] = graycoprops(glcm, 'ASM')[0, 0]
    return features
```

3.6.4 Feature Fusion

```
# Row alignment
min_len = min(len(clinical_df), len(image_df))
clinical_df =
clinical_df.iloc[:min_len].reset_index(drop=True) image_df
=
image_df.iloc[:min_len].reset_index(drop=True)

# Drop duplicate label column from clinical
clinical_features = clinical_df.drop(columns=['label'])

# Concatenate along feature axis
fused_df = pd.concat([clinical_features, image_df], axis=1)

# Remove any NaN rows
fused_df =
fused_df.dropna()
```

```
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=0.30, stratify=y, random_state=42
)
```

```
X_val, X_test, y_val, y_test = train_test_split(
```

3.6.5 Feature Selection

```
selector = SelectKBest(score_func=f_classif, k=300)
X_train_sel = selector.fit_transform(X_train, y_train)
X_val_sel    = selector.transform(X_val)

X_test_sel   = selector.transform(X_test)
```

3.6.6 SMOTE Class Balancing

```
smote = SMOTE(
    sampling_strategy=1.0,      # balance to 1:1 ratio
    random_state=42,
    k_neighbors=7

X_train_smote, y_train_smote = smote.fit_resample(X_train_sel, y_train)

# Before SMOTE: [872, 567]    (Non-Pregnant, Pregnant)
```

3.6.7 Feature Scaling

```
scaler = StandardScaler()

X_train_smote = scaler.fit_transform(X_train_smote) # fit on train
X_val_scaled  = scaler.transform(X_val_sel)         # apply to val
```

3.6.8 XGBoost Classifier

```
xgb_model = XGBClassifier(
    n_estimators=400,      learning_rate=0.03,
    max_depth=4,          min_child_weight=3,
    subsample=0.85,       colsample_bytree=0.80,
    reg_alpha=1.0,         reg_lambda=2.0,

    gamma=0.2,            scale_pos_weight=1.0,
    objective='binary:logistic', eval_metric='auc',
    random_state=42
)

xgb_model.fit()
```

3.6.9 Decision Threshold Optimization

```
y_val_prob = xgb_model.predict_proba(X_val_scaled)[: , 1]
```

```
best_threshold = 0.5
```

```
for t in np.arange(0.48, 0.60, 0.01):
    preds = (y_val_prob >= t).astype(int)
    acc = accuracy_score(y_val, preds)
    f1 = f1_score(y_val, preds)
    rec = recall_score(y_val, preds)
    score = (0.4 * acc) + (0.4 * f1) + (0.2 * rec)
    if score > best_score:
        best_score = score
        best_threshold = t
```

```
# Result: Best Threshold = 0.52
```

```
for t in np.arange(0.48, 0.60, 0.01):
    preds = (y_val_prob >= t).astype(int)
    acc = accuracy_score(y_val, preds)
    f1 = f1_score(y_val, preds)
    rec = recall_score(y_val, preds)
    score = (0.4 * acc) + (0.4 * f1) + (0.2 * rec)
    if score > best_score:
        best_score = score
        best_threshold = t
```

```
# Result: Best Threshold = 0.52
```

3.1 Model Artifacts and Deployment

```
joblib.dump(
    {
        'model'      : xgb_model,
        'scaler'     : scaler,
        'threshold'  : best_threshold
```



STUDENTS PROFILE

1	Student Name	VIBHA I S	
	USN	1BY23EC122	
	Contact No.	9483018150	
	Email id:	vibha92005@gmail.com	
	LinkedIn profile link	www.linkedin.com/in/vibha-is	
2	Student Name	ASHITHA M	
	USN	1BY23EC024	
	Contact No.	8431750512	
	Email id:	ashitha0207@gmail.com	
	LinkedIn profile link	linkedin.com/in/ashitha-m	
3	Student Name	AKSHAYA RAMESH	
	USN	1BY23EC009	
	Contact No.	8217477535	
	Email id:	akshayaramesh249@gmail.com	
	LinkedIn profile link	https://www.linkedin.com/in/akshaya-ramesh-084178326	
4	Student Name	SIREESHA T S	
	USN	1BY23EC105	
	Contact No.	8867758679	
	Email id:	sireeshats1105@gmail.com	
	LinkedIn profile link	www.linkedin.com/in/sireesha-ts-232ab2313	

TYPE OF PROJECT

Simulation		Hardware		Application (societal and environmental)		Simulation + Hardware		Product	
-------------------	--	-----------------	--	---	--	----------------------------------	--	----------------	--